

Carcinogenic potencies of polycyclic aromatic hydrocarbons for back-door neighbors of restaurants with cooking emissions.

.

In the present study, 21 polycyclic aromatic hydrocarbon (PAH) congeners were measured in the exhaust stack of 3 types of restaurants: 9 Chinese, 7 Western, and 4 barbeque (BBQ). The total PAH concentration of BBQ restaurants ($58.81 \pm 23.89 \mu\text{g m}^{-3}$) was significantly higher than that of Chinese ($20.99 \pm 13.67 \mu\text{g m}^{-3}$) and Western ($21.47 \pm 11.44 \mu\text{g m}^{-3}$) restaurants. The total benzo[a]pyrene potency equivalent (B[a]P(eq)) concentrations, however, were highest in Chinese restaurants ($1.82 \pm 2.24 \mu\text{g m}^{-3}$), followed by Western ($0.86 \pm 1.43 \mu\text{g m}^{-3}$, $p < 0.01$) and BBQ-type restaurants ($0.59 \pm 0.55 \mu\text{g m}^{-3}$, $p < 0.01$). We further developed a probabilistic risk model to assess the incremental lifetime cancer risk (ILCR) for people exposed to carcinogenic PAHs. Because the exhaust stack directly affected the back-door neighbors of these restaurants, we were concerned with the real exposure of groups near the exhaust stack outlets of these restaurants. The ILCRs for total exposure of the neighbors (inhalation+dermal contact+ingestion) were $2.6\text{--}31.3$, $1.5\text{--}14.8$, and $1.3\text{--}12.2 \times 10^{-6}$ in Chinese, Western, and BBQ restaurants, respectively. We suggest that the maximum acceptable exposure time to the exhaust stack outlet area for Chinese, Western, and BBQ restaurants ranges between 5-19, 17-42, and 18-56 h month⁻¹, respectively, based on an ILCR of less than 10^{-6} . Crown Copyright © 2011. Published by Elsevier B.V. All rights reserved.